

Lab 8

Practice 1

```
1: thecoon@thecoon: ~/Documents
thecoon@thecoon:~/Documents$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lpr:x:7:7:/usr/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:11:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
lirc:x:39:39:lirc:/run/lirc:/usr/sbin/nologin
_apt:x:42:65534:/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
system-network:x:998:998:system Network Management:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
tss:x:101:103:TPM software stack:/var/lib/tpm:/bin/false
system-timesync:x:991:991:system Time Synchronization:/usr/sbin/nologin
messagebus:x:990:990:system Message Bus:/nonexistent:/usr/sbin/nologin
dmsquash:x:999:65534:dmsquash:/var/lib/misc:/usr/sbin/nologin
avahi:x:182:187:avahi mDNS daemon:/run/avahi-daemon:/usr/sbin/nologin
speech-dispatcher:x:183:29:speech-dispatcher:/run/speech-dispatcher:/bin/false
usbmux:x:104:46:usbmux daemon:/var/lib/usbmux:/usr/sbin/nologin
cups-pk-helper:x:105:108:user for cups-pk-helper services:/nonexistent:/usr/sbin/nologin
fwupd-refresh:x:989:989:firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
geoclue:x:106:118:/var/lib/geoclue:/usr/sbin/nologin
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
saned:x:107:111:/var/lib/saned:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/usr/sbin/nologin
rtkit:x:108:112:RealtimeKit:/proc:/usr/sbin/nologin
colord:x:109:113:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
deban-gdm:x:110:114:Gnome Display Manager:/var/lib/gdm:/bin/false
thecoon:1000:1000:thecoon,,/home/thecoon:/bin/bash
_flatpak:x:111:115:Flatpak system-wide installation helper:/nonexistent:/usr/sbin/nologin
vboxadd:x:997:1:/var/run/vboxadd:/bin/false
thecoon@thecoon:~/Documents$ tail -n 5 /etc/passwd
vboxadd:x:997:1:/var/run/vboxadd:/bin/false
thecoon:1000:1000:thecoon,,/home/thecoon:/bin/bash
deban-gdm:x:110:114:Gnome Display Manager:/var/lib/gdm:/bin/false
colord:x:109:113:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
rtkit:x:108:112:RealtimeKit:/proc:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/usr/sbin/nologin
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
geoclue:x:106:118:/var/lib/geoclue:/usr/sbin/nologin
fwupd-refresh:x:989:989:firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin

1: thecoon@thecoon: ~/Documents
thecoon@thecoon:~/Documents$ nl -ba /etc/passwd | cat -E
 1 root:x:0:0:root:/root:/bin/bash$
 2 daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin$
 3 bin:x:2:2:bin:/bin:/usr/sbin/nologin$
 4 sys:x:3:3:sys:/dev:/usr/sbin/nologin$
 5 sync:x:4:65534:sync:/bin:/bin/sync$
 6 games:x:5:60:games:/usr/games:/usr/sbin/nologin$
 7 man:x:6:12:man:/var/cache/man:/usr/sbin/nologin$
 8 lpr:x:7:7:lp:/usr/spool/lpd:/usr/sbin/nologin$
 9 mail:x:8:8:mail:/var/mail:/usr/sbin/nologin$
10 news:x:9:9:news:/var/spool/news:/usr/sbin/nologin$
11 uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin$
12 proxy:x:11:13:proxy:/bin:/usr/sbin/nologin$
13 www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin$
14 backup:x:34:34:backup:/var/backups:/usr/sbin/nologin$
15 lirc:x:39:39:lirc:/run/lirc:/usr/sbin/nologin$
16 _apt:x:42:65534:/nonexistent:/usr/sbin/nologin$
17 nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin$
18 system-network:x:998:998:system Network Management:/usr/sbin/nologin$
19 dhcpcd:x:100:65534:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false$
20 tss:x:101:103:TPM software stack:/var/lib/tpm:/bin/false$
21 system-timesync:x:991:991:system Time Synchronization:/usr/sbin/nologin$
22 messagebus:x:990:990:system Message Bus:/nonexistent:/usr/sbin/nologin$
23 dmsquash:x:999:65534:dmsquash:/var/lib/misc:/usr/sbin/nologin$
24 avahi:x:182:187:avahi mDNS daemon:/run/avahi-daemon:/usr/sbin/nologin$
25 speech-dispatcher:x:183:29:speech-dispatcher:/run/speech-dispatcher:/bin/false$
26 usbmux:x:104:46:usbmux daemon:/var/lib/usbmux:/usr/sbin/nologin$
27 cups-pk-helper:x:105:108:user for cups-pk-helper services:/nonexistent:/usr/sbin/nologin$
28 fwupd-refresh:x:989:989:firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin$
29 geoclue:x:106:118:/var/lib/geoclue:/usr/sbin/nologin$
30 gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin$
31 saned:x:107:111:/var/lib/saned:/usr/sbin/nologin$
32 polkitd:x:987:987:User for polkitd:/usr/sbin/nologin$
33 rtkit:x:108:112:RealtimeKit:/proc:/usr/sbin/nologin$
34 colord:x:109:113:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin$
35 deban-gdm:x:110:114:Gnome Display Manager:/var/lib/gdm:/bin/false$
36 thecoon:x:1000:1000:thecoon,,/home/thecoon:/bin/bash$
37 _flatpak:x:111:115:Flatpak system-wide installation helper:/nonexistent:/usr/sbin/nologin$
38 vboxadd:x:997:1:/var/run/vboxadd:/bin/false$
thecoon@thecoon:~/Documents$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
thecoon@thecoon:~/Documents$ tail -n 5 /etc/passwd
vboxadd:x:997:1:/var/run/vboxadd:/bin/false
thecoon:1000:1000:thecoon,,/home/thecoon:/bin/bash
deban-gdm:x:110:114:Gnome Display Manager:/var/lib/gdm:/bin/false
colord:x:109:113:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
rtkit:x:108:112:RealtimeKit:/proc:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/usr/sbin/nologin
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
geoclue:x:106:118:/var/lib/geoclue:/usr/sbin/nologin
fwupd-refresh:x:989:989:firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin

1: thecoon@thecoon: ~/Documents
thecoon@thecoon:~/Documents$ wc -l ~/Documents/sample_files/Text/dracula.txt
15845 ~/Documents/sample_files/Text/dracula.txt
```

Practice 2

```

Tilix: thecoon@thecoon: ~
1: thecoon@thecoon: ~
thecoon@thecoon:~$ cut -d":" -f1 /etc/passwd
root
daemon
bin
sys
sync
games
man
lp
mail
news
uucp
proxy
www-data
backup
list
irc
_apt
nobody
systemd-network
dhcpcd
tss
systemd-timesync
messagebus
dnsmasq
avahi
speech-dispatcher
usbmux
cups-pk-helper
fwupd-refresh
geoclue
gnome-remote-desktop
saned
polkitd
rtkit
colord
Debian-gdm
thecoon
_flatpak
vboxadd
thecoon@thecoon:~$ cut -d":" -f1,6 /etc/passwd
root:/root
daemon:/usr/sbin
bin:/bin
sys:/dev
sync:/bin
games:/usr/games
man:/var/cache/man
lp:/var/spool/lpd
mail:/var/mail
news:/var/spool/news
uucp:/var/spool/uucp
proxy:/bin
www-data:/var/www
backup:/var/backups
list:/var/list
irc:/run/ircd
_apt:/nonexistent
nobody:/nonexistent
systemd-network:/
dhcpcd:/usr/lib/dhcpcd
tss:/var/lib/tpm
systemd-timesync:/
messagebus:/nonexistent
dnsmasq:/var/lib/misc
avahi:/run/avahi-daemon
speech-dispatcher:/run/speech-dispatcher
usbmux:/var/lib/usbmux
cups-pk-helper:/nonexistent
fwupd-refresh:/var/lib/fwupd
geoclue:/var/lib/geoclue
gnome-remote-desktop:/var/lib/gnome-remote-desktop
saned:/var/lib/saned
polkitd:/
rtkit:/proc
colord:/var/lib/colord
Debian-gdm:/var/lib/gdm3
thecoon:/home/thecoon
_flatpak:/nonexistent
vboxadd:/var/run/vboxadd
thecoon@thecoon:~$ cut -d":" -f1,6,7 --output-delimiter=" " /etc/passwd
root /root /bin/bash
daemon /usr/sbin /usr/sbin/nologin
bin /bin /usr/sbin/nologin
sys /dev /usr/sbin/nologin
sync /bin /bin/sync
games /usr/games /usr/sbin/nologin
man /var/cache/man /usr/sbin/nologin
lp /var/spool/lpd /usr/sbin/nologin
mail /var/mail /usr/sbin/nologin
news /var/spool/news /usr/sbin/nologin
uucp /var/spool/uucp /usr/sbin/nologin
proxy /bin /usr/sbin/nologin
www-data /var/www /usr/sbin/nologin
backup /var/backups /usr/sbin/nologin
list /var/list /usr/sbin/nologin
irc /run/ircd /usr/sbin/nologin
_apt /nonexistent /usr/sbin/nologin
nobody /nonexistent /usr/sbin/nologin
systemd-network / /usr/sbin/nologin
systemd-network / /usr/sbin/nologin
dhcpcd /usr/lib/dhcpcd /bin/false
tss /var/lib/tpm /bin/false
systemd-timesync / /usr/sbin/nologin
messagebus /nonexistent /usr/sbin/nologin
dnsmasq /var/lib/misc /usr/sbin/nologin
avahi /run/avahi-daemon /usr/sbin/nologin
speech-dispatcher /run/speech-dispatcher /bin/false
usbmux /var/lib/usbmux /usr/sbin/nologin
cups-pk-helper /nonexistent /usr/sbin/nologin
fwupd-refresh /var/lib/fwupd /usr/sbin/nologin
geoclue /var/lib/geoclue /usr/sbin/nologin
gnome-remote-desktop /var/lib/gnome-remote-desktop /usr/sbin/nologin
saned /var/lib/saned /usr/sbin/nologin
polkitd / /usr/sbin/nologin
rtkit /proc /usr/sbin/nologin
colord /var/lib/colord /usr/sbin/nologin
Debian-gdm /var/lib/gdm3 /bin/false
thecoon /home/thecoon /bin/bash
_flatpak /nonexistent /usr/sbin/nologin
vboxadd /var/run/vboxadd /bin/false
thecoon@thecoon:~$ cut -d":" -f1 --output-delimiter=" " /etc/passwd
name
String
100% Bran
100% Natural Bran
11.8 g

```

```
1: thecoon@thecoon: ~  
thecoon@thecoon:~$ cut -d";" -f1 ~/Documents/sample_files/Csv/cereal.csv  
name  
String  
100% Bran  
100% Natural Bran  
All-Bran  
All-Bran with Extra Fiber  
Almond Delight  
Apple Cinnamon Cheerios  
Apple Jacks  
Basic 4  
Bran Chex  
Bran Flakes  
Cap'n Crunch  
Cheerios  
Cinnamon Toast Crunch  
Clusters  
Cocoa Puffs  
Corn Chex  
Corn Flakes  
Corn Pops  
Count Chocula  
Cracklin' Oat Bran  
Cream of Wheat (Quick)  
Crispix  
Crispy Wheat & Raisins  
Double Chex  
Froot Loops  
Frosted Flakes  
Frosted Mini-Wheats  
Fruit & Fibre Dates, Walnuts, and Oats  
Fruitful Bran  
Fruity Pebbles  
Golden Crisp  
Golden Grahams  
Grape Nuts Flakes  
Grape-Nuts  
Great Grains Pecan  
Honey Graham Ohs  
Honey Nut Cheerios  
Honey-comb  
Just Right Crunchy Nuggets  
Just Right Fruit & Nut  
Kix  
Life  
Lucky Charms
```

```
Applications May 1 7:27 PM  
1: thecoon@thecoon: ~  
thecoon@thecoon:~$ cut -d";" -f1,4 --output-delimiter" calories = " ~/Documents/sample_files/Csv/cereal.csv  
name calories = calories  
String calories = int  
100% Bran calories = 70  
100% Natural Bran calories = 120  
All-Bran calories = 70  
All-Bran with Extra Fiber calories = 50  
Almond Delight calories = 110  
Apple Cinnamon Cheerios calories = 110  
Apple Jacks calories = 110  
Basic 4 calories = 130  
Bran Chex calories = 90  
Bran Flakes calories = 90  
Cap'n Crunch calories = 120  
Cheerios calories = 110  
Cinnamon Toast Crunch calories = 120  
Clusters calories = 110  
Cocoa Puffs calories = 110  
Corn Chex calories = 110  
Corn Flakes calories = 200  
Corn Pops calories = 110  
Count Chocula calories = 110  
Cracklin' Oat Bran calories = 110  
Cream of Wheat (Quick) calories = 100  
Crispix calories = 110  
Crispy Wheat & Raisins calories = 100  
Double Chex calories = 100  
Froot Loops calories = 110  
Frosted Flakes calories = 110  
Frosted Mini-Wheats calories = 200  
Fruit & Fibre Dates, Walnuts, and Oats calories = 120  
Fruitful Bran calories = 120  
Fruity Pebbles calories = 110  
Golden Crisp calories = 100  
Golden Grahams calories = 110  
Grape Nuts Flakes calories = 100  
Grape-Nuts calories = 110  
Great Grains Pecan calories = 120  
Honey Graham Ohs calories = 120  
Honey Nut Cheerios calories = 110  
Honey-comb calories = 110  
Just Right Crunchy Nuggets calories = 110  
Just Right Fruit & Nut calories = 140  
Kix calories = 110  
Life calories = 100  
Lucky Charms calories = 110  
M&M's calories = 100  
M&M's Raisins, Dates, & Almonds calories = 150  
M&M's Raisins, Peaches, & Pecans calories = 150  
M&M's Crispy Blend calories = 160  
Multi-Grain Cheerios calories = 100  
NutriHoney Crunch calories = 120
```

Practice 3

```
1: thecoon@thecoon: ~/Documents/sample_files/Csv  
thecoon@thecoon:~$ sort ~/Documents/sample_files/Csv/cars.csv > ~/Documents/sample_files/Csv/sorted_cars.csv  
thecoon@thecoon:~$ cd ~/Documents/sample_files/Csv/  
thecoon@thecoon:~/Documents/sample_files/Csv$ wc -l sorted_cars.csv  
408 sorted_cars.csv  
thecoon@thecoon:~/Documents/sample_files/Csv$ wc -m sorted_cars.csv  
22063 sorted_cars.csv  
thecoon@thecoon:~/Documents/sample_files/Csv$ sed 's/;/,/g' sorted_cars.csv > sorted_cars_commas.csv  
thecoon@thecoon:~/Documents/sample_files/Csv$
```

Question 2

```
Tilix: thecoon@thecoon: ~  
1: thecoon@thecoon: ~  
thecoon@thecoon:~$ curl -s https://cis106.com/assets/scripts/cha18.sh | bash  
[INFO] Generating 100 log entries in access.log...  
[SUCCESS] Access.log created successfully. Location: /home/thecoon/access.log  
thecoon@thecoon:~$ wc -l access.log  
100 access.log  
thecoon@thecoon:~$ head -n 1 access.log  
2025-10-13 12:00:12 Asimov /admin DENIED  
thecoon@thecoon:~$ tail -n 5 access.log  
2025-11-05 08:11:06 Rize /download SUCCESS  
thecoon@thecoon:~$ head -n 15 access.log  
2025-10-13 12:00:12 Asimov /admin DENIED  
2025-10-13 15:47:41 Rize /download DENIED  
2025-07-18 21:51:16 Rei /dashboard DENIED  
2025-08-20 16:21:59 Eren /download FAIL  
2025-07-20 07:05:44 Spike /upload FAIL  
2025-01-24 02:40:43 Faye /404 DENIED  
2025-09-11 05:41:40 Edward /logout SUCCESS  
2025-04-20 22:21:22 Ein /upload FAIL  
2025-11-18 23:14:29 Honami /login DENIED  
2025-02-25 14:42:38 Kei /admin FAIL  
2025-04-10 09:59:51 Ken /404 DENIED  
2025-12-02 19:25:22 Haruka /login SUCCESS  
2025-09-17 09:31:15 Faye /dashboard FAIL  
2025-02-23 17:17:27 Jet /admin DENIED  
2025-03-12 13:20:16 Jet /login SUCCESS  
thecoon@thecoon:~$ tail -n 15 access.log  
2025-05-12 10:55:52 Honami /admin DENIED  
2025-02-20 12:10:35 Edward /dashboard FAIL  
2025-08-14 13:25:40 Eren /dashboard SUCCESS  
2025-12-27 22:06:05 Asimov /settings SUCCESS  
2025-12-04 19:33:24 Honami /admin FAIL  
2025-12-25 01:35:24 Walter /upload FAIL  
2025-10-14 10:05:24 Walter /reset FAIL  
2025-09-11 13:59:02 Haruka /dashboard FAIL  
2025-07-14 13:02:23 Seras /logout SUCCESS  
2025-04-09 12:54:54 Kei /logout FAIL  
2025-04-03 07:42:56 Seras /dashboard DENIED  
2025-06-22 19:09:50 Walter /login FAIL  
2025-03-01 16:40:20 Arisu /download DENIED  
2025-12-15 16:04:21 Ken /login SUCCESS  
2025-11-05 08:11:06 Rize /download SUCCESS  
thecoon@thecoon:~$ cut -d' ' -f3 access.log  
Asimov  
Rize  
Rei  
Eren  
Spike  
Faye  
Edward  
Ein  
Honami  
Kei  
Ken  
Haruka  
Faye  
Jet  
Jet  
Shinji  
Rize  
Arisu  
Ken  
Edward  
Rize  
Ken  
Shinji  
Arisu  
Julia  
Tonka  
Asimov  
Ken  
Ken  
Faye  
Walter  
Ken  
Jet  
Ken  
Arisu  
Seras  
Seras  
Shinji  
Haruka  
Arisu  
Shinji
```

```
Tilix: thecoon@thecoon: ~  
1: thecoon@thecoon: ~  
thecoon@thecoon:~$ sort -k3 access.log -o sorted_access_log_by_uname.log  
thecoon@thecoon:~$ sort -k4 access.log -o sorted_access_log_by_endpoint.log  
thecoon@thecoon:~$ cut -d' ' -f3 access.log  
Asimov  
Rize  
Rei  
Eren  
Spike  
Faye  
Edward  
Ein  
Honami  
Kei  
Ken  
Haruka  
Faye  
Jet  
Jet  
Shinji  
Rize  
Arisu  
Ken  
Edward  
Rize  
Ken  
Shinji  
Arisu  
Julia  
Tonka  
Asimov  
Ken  
Ken  
Faye  
Walter  
Ken  
Jet  
Ken  
Arisu  
Seras  
Seras  
Shinji  
Haruka  
Arisu  
Shinji
```

```
Tilix: thecoon@thecoon: ~  
1: thecoon@thecoon: ~  
thecoon@thecoon:~$ cut -d' ' -f4 access.log  
/admin  
/download  
/dashboard  
/download  
/upload  
/404  
/logout  
/upload  
/login  
/admin  
/404  
/login  
/dashboard  
/admin  
/login  
/logout  
/settings  
/download  
/download  
/download  
/download  
/download  
/upload  
/dashboard  
/reset  
/reset  
/reset  
/login  
/404  
/404  
/upload  
/logout  
/reset  
/reset  
/reset  
/404  
/login  
/upload  
/settings  
/dashboard  
/reset  
/download  
/logout  
/logout  
/:
```